Comparing Thalamic and Cortical Activity in Visual Processing

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1 Introduction

The thalamus and cortex are key brain regions involved in processing visual information. While the thalamus acts as a relay station, transmitting signals from the retina to the cortex, the cortical areas perform higher level processing of visual input. Understanding how neural activity differs between these regions can provide insights into their distinct roles and the flow of visual information in the brain. This project aims to compare thalamic and cortical activity in mice exposed to visual stimuli, using data from the Allen Brain Observatory.

2 Research Question

How does thalamic activity compare to cortical activity when processing visual input?

3 Methodology

The analysis was conducted using a Jupyter notebook containing code, outputs, and commentary for analyzing Allen Brain Observatory data on thalamic and cortical activity in response to visual stimuli. The main steps included:

1. Accessing electrophysiology data for a session with a substantial number of units recorded in both thalamic (LGd, LP, PO) and cortical (VISp, VISl, VISal, VISpm, VISam, VISrl) regions.
2. Extracting and comparing neural activity (spike times, firing rates) between thalamic and cortical units in response to various visual stimuli, with a focus on natural scenes.
3. Visualizing and statistically comparing spike count distributions, firing rate distributions, and mean firing rates between the two regions.
4. Examining temporal dynamics and performing principal component analysis (PCA) and logistic regression classification to assess differences in thalamic vs cortical responses.

4 Results

Key findings from the analysis include:

- Thalamic units exhibited significantly higher mean firing rates compared to cortical units in response to visual stimuli (thalamic: 9.38 Hz, cortical: 6.44 Hz, $p < 0.001$). This difference was consistent across stimulus driven responses.
- Spike count distributions showed higher average spike counts per stimulus presentation in thalamic neurons compared to cortical neurons.
- PCA revealed some separation between thalamic and cortical populations based on their activity patterns, but with substantial overlap. A logistic regression classifier was only moderately successful in distinguishing the two regions based on firing rates alone (accuracy 59%).

- Temporal response profiles suggested thalamic neurons may respond earlier however there is still significant overlap in the distribution.

5 Discussion

The analysis shows that mean firing rates are significantly higher in thalamic units compared with cortical units. In the visual pathway, thalamic nuclei such as LGd, LP, and PO serve primarily as relay stations from the retina to the cortex. They tend to "fire fast" and more reliably to preserve the timing and contrast of incoming visual signals. Since they receive direct retinal input, their responses tend to be more homogeneous, which is reflected in their spike count statistics. In contrast, neurons in cortical areas (VISp, VISl, etc.) integrate their inputs, and are involved in more complex feature extraction. Thus, the statistically significant difference reflects how thalamic neurons are optimized for rapid transmission, whereas cortical neurons are engaged in more elaborate computations that might involve lower average firing rates or more selective activation.

When comparing responses to different stimuli (e.g., "natural_scenes" versus "spontaneous"), the notebook shows that the overall spike count statistics differ markedly. The thalamus, with its more "hard wired" relay properties, may maintain a certain level of baseline and stimulus driven firing even in spontaneous conditions. Cortical neurons, which integrate context, attention, and ongoing cognitive state, could display more variability.

A simple logistic regression classifier based solely on firing rate achieved an accuracy of 58.7%, suggesting that although the average rate is somewhat discriminative, there is still considerable overlap between the two populations. This moderate classification performance is in line with the idea that while thalamic neurons are on average more active, many cortical neurons (depending on the particular visual stimulus or behavioral state) can display overlapping firing ranges.

In contrast, when the the data was first reduced in dimension via PCA and then classified with logistic regression, the accuracy dropped further (to roughly 46.8%). PCA extracted the principal directions of variance in the data, those dimensions did not necessarily align with the aspects that best separate thalamic from cortical responses. This lower accuracy implies that the global patterns of trial spike counts in these brain regions are not easily linearly separable in the reduced space.

The moderate accuracy of classifiers (especially the PCA based method) hints that more complex or nonlinear models might be necessary to capture the diversity of response profiles in these regions. Future work could use more sophisticated feature extraction (e.g., nonlinear dimensionality reduction, spike pattern analysis) to better identify discriminative patterns (which was not done for the sake of computational time).

It would also be valuable to further examine stimulus specificity further, for example, whether thalamic neurons remain responsive across a larger variety of visual stimuli while cortical neurons differentiate features such as orientation, motion, or color.

References

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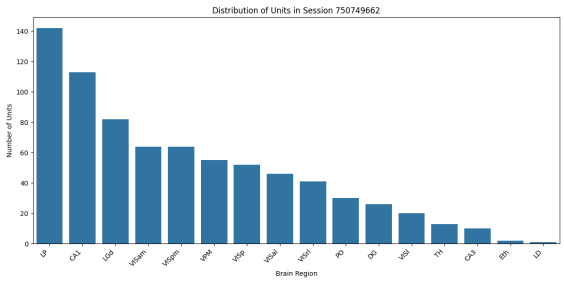
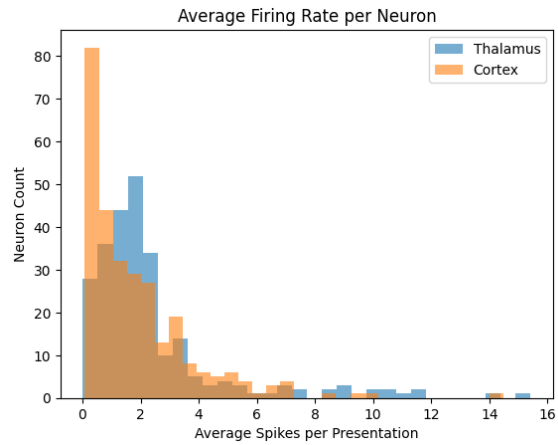
[2] D. H. Hubel and T. N. Wiesel. "Receptive fields, binocular interaction and functional architecture in the cat's visual cortex". In: *The Journal of Physiology* (1962).

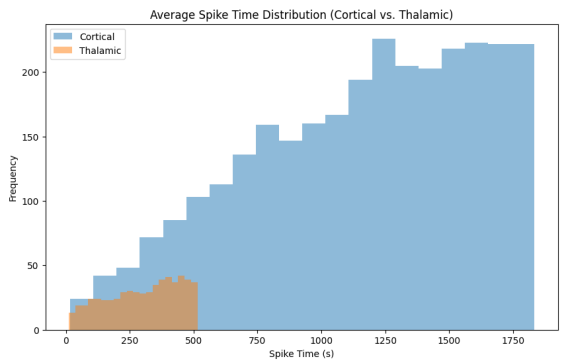
[3] R. J. Douglas and K. A. C. Martin. "Neuronal circuits of the neocortex". In: *Annual Review of Neuroscience* (2004).



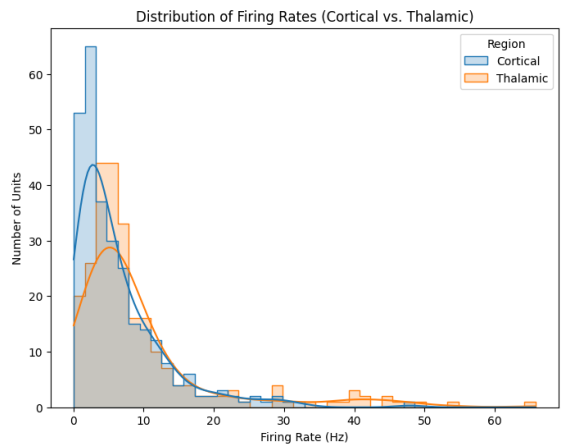
(a) Brain Regions Distributions



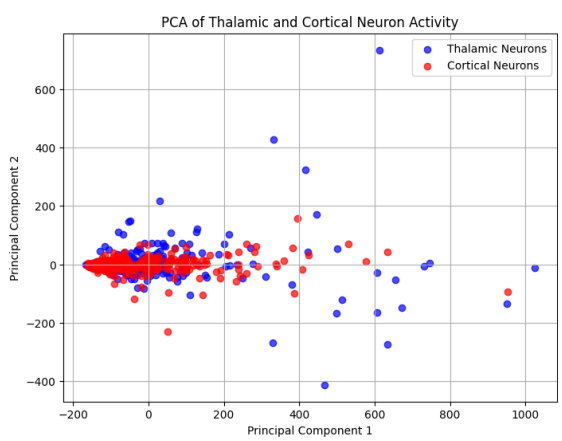
(b) Average Spikes per presentation: Cortex vs Thalamus



(c) Average Spike Time Distribution: Cortical vs. Thalamic



(d) Firing Rate: Corrtical vs Thalamic



(e) PCA